

# Stage 3 Judges' Rubric - DEMI Grade 6

Robotics + AI Project - Stage 3 Judge Evaluation

|                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|
| Team Name / #        | Project Name         | Judge Name           | Date                 |
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |

**How to Score Every Row** Enter ONE number from 1 to 5. Every row is worth 5 points; harder sections have more rows and therefore more weight.

|   |                    |                                                                                                  |
|---|--------------------|--------------------------------------------------------------------------------------------------|
| 1 | Weak               | Missing, incorrect, non-functional, or little/no meaningful evidence.                            |
| 2 | Developing         | Partially demonstrated; important gaps remain or substantial support is needed.                  |
| 3 | Meets Expectations | Core requirement is present, understandable, and works at an acceptable/basic level.             |
| 4 | Strong             | Complete, accurate, well integrated, and supported by good evidence; only minor issues.          |
| 5 | Excellent          | Highly effective, reliable, purposeful, well-tested/polished, and clearly justified by the team. |

**Judge the final Stage 3 work, using the previous-phase requirements as the baseline.** **Flow:** reward evidence, correct integration, and purposeful design - not complexity for its own sake. **Testing:** look for real trials/results, not only claims. **Hardware:** reward the best-fit use of components and reliable integration. **Teamwork & presentation:** look for shared ownership, clear communication, and the ability of multiple members to answer questions.

| Specific Judging Task             | What the Judge Should Look For                                                                                         | Score (1-5)               | Judge Note           |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------|
| Problem, Research & Solution      |                                                                                                                        | Section Weight: 15 points |                      |
| Problem, user & research          | Problem and target user are specific, important, and supported by relevant research or comparison with existing ideas. | <input type="text"/>      | <input type="text"/> |
| Solution fit                      | Main functions clearly solve the identified problem and remain consistent with the same user and use case.             | <input type="text"/>      | <input type="text"/> |
| Value & overall concept coherence | Hardware, programming, AI, robotics, and user value form one coherent solution rather than separate features.          | <input type="text"/>      | <input type="text"/> |
| Hardware, System & Robotics       |                                                                                                                        | Section Weight: 30 points |                      |
| Purposeful hardware selection     | Kit, sensors, servos/actuators, and other parts are chosen because they best support the solution.                     | <input type="text"/>      | <input type="text"/> |

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| Specific Judging Task                   | What the Judge Should Look For                                                                                 | Score (1-5)          | Judge Note                |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|
| Hardware, System & Robotics (continued) |                                                                                                                |                      | Section Weight: 30 points |
| Input/output roles                      | Sensor inputs and actuator/servo outputs are technically correct and clearly explained.                        | <input type="text"/> | <input type="text"/>      |
| Circuit & system design                 | Circuit and block/system design are correct, organized, and match the implemented prototype.                   | <input type="text"/> | <input type="text"/>      |
| Robot movement sequence                 | Robot movement/behavior is logical, coordinated, and matches the planned sequence or task.                     | <input type="text"/> | <input type="text"/>      |
| Hardware integration & control          | Arduino/controller reliably connects sensing, decisions, and physical movement as one system.                  | <input type="text"/> | <input type="text"/>      |
| Best hardware use & reliability         | Team uses available hardware capabilities effectively and demonstrates consistent, safe, repeatable operation. | <input type="text"/> | <input type="text"/>      |

|                                    |                                                                                                                  |                      |                           |
|------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|
| Programming & AI                   |                                                                                                                  |                      | Section Weight: 20 points |
| Core programming logic             | Events, variables, conditions, loops, and outputs correctly control the system.                                  | <input type="text"/> | <input type="text"/>      |
| Functions / lists / stronger logic | Uses functions, lists, or other suitable concepts to organize and strengthen the program.                        | <input type="text"/> | <input type="text"/>      |
| AI/ML approach & relevance         | AI/ML type (classification, regression, clustering, etc.) is appropriate and meaningfully supports the solution. | <input type="text"/> | <input type="text"/>      |
| AI evidence & system connection    | Shows evidence of the AI concept/model and explains how its decision connects to the robot/system behavior.      | <input type="text"/> | <input type="text"/>      |

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| Specific Judging Task                 | What the Judge Should Look For                                                                                        | Score (1-5)          | Judge Note                |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------|
| Prototype & Testing                   |                                                                                                                       |                      | Section Weight: 15 points |
| End-to-end prototype                  | Core sensing, decision, programming, and robot behavior work together in the final demonstration.                     | <input type="text"/> | <input type="text"/>      |
| Evidence of testing                   | Team provides repeated trials, measurements, logs, videos/screenshots, or success counts that prove testing occurred. | <input type="text"/> | <input type="text"/>      |
| Improvement from results              | Testing leads to meaningful changes in thresholds, movement, timing, model behavior, or reliability.                  | <input type="text"/> | <input type="text"/>      |
| Business & Value                      |                                                                                                                       |                      | Section Weight: 5 points  |
| Target customer & value               | Business idea identifies the user/customer and explains the practical value of the solution.                          | <input type="text"/> | <input type="text"/>      |
| Teamwork & Presentation               |                                                                                                                       |                      | Section Weight: 15 points |
| Shared ownership & teamwork           | Members have clear roles, contribute meaningfully, and support one another during build/demo.                         | <input type="text"/> | <input type="text"/>      |
| Communication & technical explanation | Team explains the robot, hardware, code, AI, and testing clearly and accurately.                                      | <input type="text"/> | <input type="text"/>      |
| Presentation, demo & Q&A;             | Presentation is structured, confident, and within time; questions are answered with evidence and understanding.       | <input type="text"/> | <input type="text"/>      |

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| Specific Judging Task               | What the Judge Should Look For                                                                                                   | Score (1-5)          | Judge Note              |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------|
| Extra Requirements / Optional Bonus |                                                                                                                                  |                      | Optional Bonus - max 15 |
| Advanced programming feature        | Optional advanced code feature meaningfully improves the existing system and is supported by code/logic evidence.                | <input type="text"/> | <input type="text"/>    |
| Advanced AI/ML feature              | Optional AI/ML feature goes deeper than the core AI requirement and is supported by evidence or a credible prototype.            | <input type="text"/> | <input type="text"/>    |
| Godot application                   | Optional Godot feature meaningfully explains, simulates, or supports the same project concept and includes design/code evidence. | <input type="text"/> | <input type="text"/>    |

## Score Summary

| Section                      | Maximum | Section Subtotal     | Notes                |
|------------------------------|---------|----------------------|----------------------|
| Problem, Research & Solution | 15      | <input type="text"/> | <input type="text"/> |
| Hardware, System & Robotics  | 30      | <input type="text"/> | <input type="text"/> |
| Programming & AI             | 20      | <input type="text"/> | <input type="text"/> |
| Prototype & Testing          | 15      | <input type="text"/> | <input type="text"/> |
| Business & Value             | 5       | <input type="text"/> | <input type="text"/> |
| Teamwork & Presentation      | 15      | <input type="text"/> | <input type="text"/> |
| CORE TOTAL                   | 100     | <input type="text"/> |                      |

|                                            |                      |
|--------------------------------------------|----------------------|
| OPTIONAL BONUS (record separately, max 15) | <input type="text"/> |
|--------------------------------------------|----------------------|